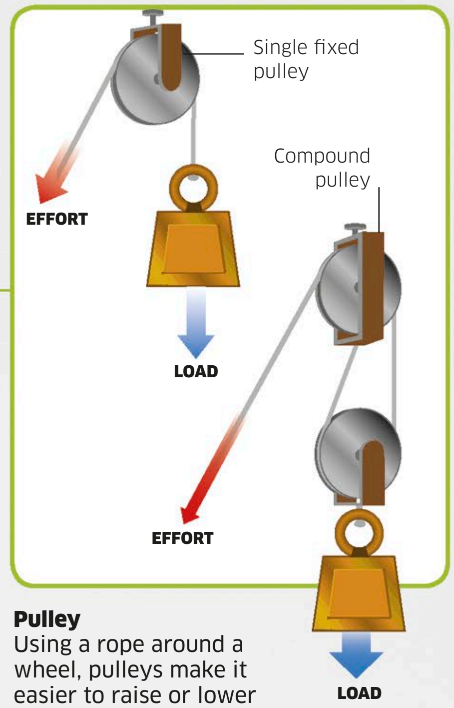
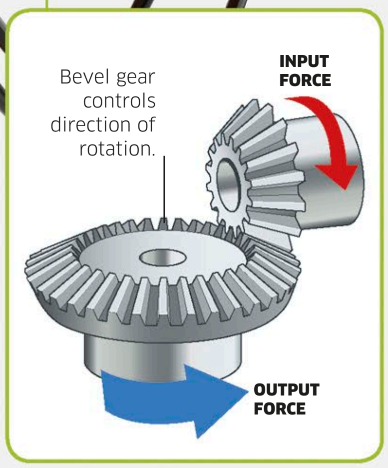
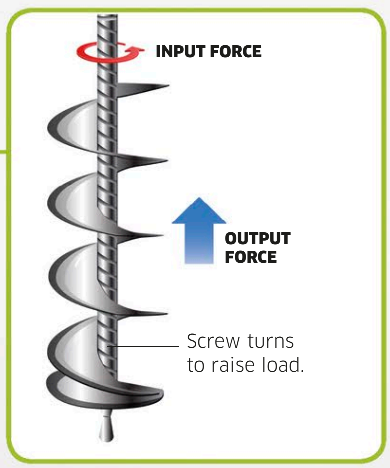


Pulley
Using a rope around a wheel, pulleys make it easier to raise or lower a load. A single fixed pulley changes the direction of movement. A compound (block and tackle) pulley reduces the effort, too.



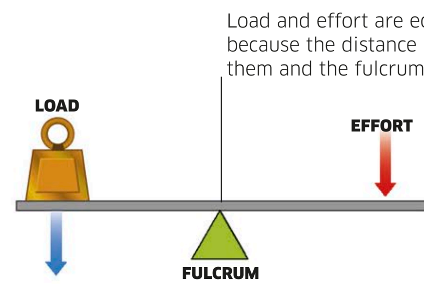
Gears
Gears are toothed wheels that transmit force and come in four main types. In all of them, one gear wheel turns faster or slower than the other or moves in a different direction. In "bevel" gears, two wheels interlock to change the direction of rotation.



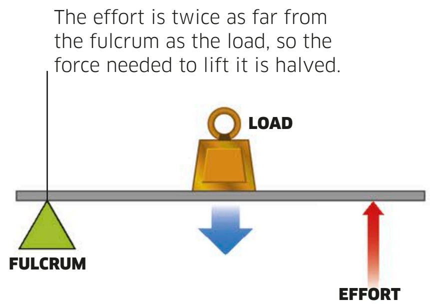
Screw
An auger—the screwlike drill bit of this boring tool—is a ramp that winds around itself, with a wedge at the tip. It is used to lift earth as it excavates. Other screws, such as light bulbs or wood and masonry screws, hold things together.

Types of lever

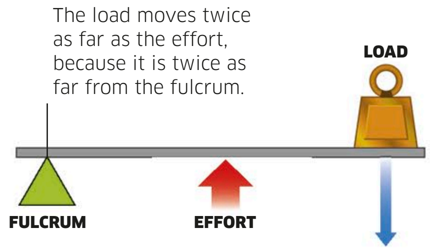
A lever is a bar that tilts on a fulcrum or pivot. If you apply force (effort) to one part of a lever, the lever swings on the fulcrum to raise a load. Levers work in three ways, depending on the relative position of the fulcrum, load, and effort on the bar.



First-class levers
The fulcrum is in between the effort and the load—as in a beam scale or a pair of scissors (two levers hinged at a fulcrum).



Second-class levers
The fulcrum is at one end and effort is applied to the other, with a load between—as in a wheelbarrow or nutcracker.



Third-class levers
The fulcrum is at the end, with load at the other end and effort applied in between—as in a hammer or a pair of tweezers.

Compound machines

A big mechanical crane and digger combines a number of simple machines with a powerful engine to make light work of heavy lifting and excavation.